THINKING

Reasoning

deduction, induction, abduction

Problem solving

Deductive Reasoning

- Deduction:
  - derive logically necessary conclusion from given premises.

    e.g. If it is Friday then she will go to temple

    It is Friday

    Therefore she will go to temple

- Logical conclusion not necessarily true:

  e.g. If it is raining then the ground is dry

  It is raining

  Therefore the ground is dry

Deduction (cont.)

- When truth and logical validity clash …

  e.g. Some people are babies

  Some babies cry

  Inference - Some people cry

  Correct?

- People bring world knowledge to bear

Inductive Reasoning

- Induction:
  - generalize from cases seen to cases unseen

    e.g. all elephants we have seen have trunks
    therefore all elephants have trunks.

- Unreliable:
  - can only prove false not true

… but useful!

Abductive reasoning

- reasoning from event to cause

  e.g. Sam drives fast when drunk.

  If I see Sam driving fast, assume drunk.

- Unreliable:
  - can lead to false explanations

Problem solving

- Process of finding solution to unfamiliar task using knowledge
- Several theories.
  - Gestalt Theory
  - Problem space theory
- Gestalt Theory
  - Past experience affects individual perception
  - Stimuli grouped in own perception patterns
  - Processes leave a trace in the brain (encoding)
  - Group information to make it more meaningful

Gestalt - Overview

- Gestalt is a psychology term which means "unified whole".
- It refers to theories of **visual perception** developed by German psychologists in the 1920s.

Gestalt (cont.)

- These theories attempt to describe how people tend to organize visual elements into **groups** or **unified wholes** when certain **principles** are applied. These principles are:
  - Similarity
  - Continuation
  - Closure
  - Proximity
  - Figure and ground

Gestalt (cont.)

- Similarity occurs when object look similar to one another
- People often perceive them as group or pattern

- **Unity** occurs because the triangular shapes at the bottom of the eagle **look similar**

Gestalt (cont.)

- Continuation occurs when the eye is compelled to move through one object and continue to another object

- Continuation occurs here because the smooth flowing crossbar of the "H" leads the eye directly to the maple leaf

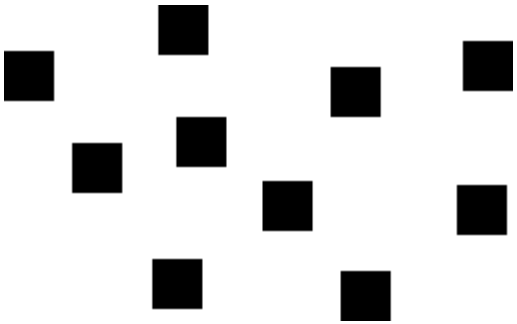
Gestalt (cont.)

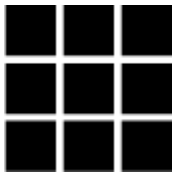
- Closure occurs when an object is incomplete
- If enough of the shape is indicated, people perceive the whole by filling in the missing information

- Closure occurs here although the panda above is not complete, enough is present for the eye to complete the shape

Gestalt (cont.)

- Proximity occurs when elements are placed close together.





- The squares on the left are placed **without proximity**. They are perceived as separate shapes.
- When the squares are given **close proximity**, unity occurs, they are now perceived as one group.

Gestalt (cont.)

- Figure
  - That what is the important subject
  - Foreground
- Ground
  - That what is the environment of the subject
  - Background
- The eye actively separate Figure from Ground
  - Selecting what is important
- This image uses complex figure/ground relationships which change upon perceiving leaves, water and tree trunk.

Problem solving (cont.)

Problem space theory

- problem space comprises problem states
- problem solving involves generating states using legal operators
- heuristics may be employed to select operators
  - e.g. means-ends analysis
- operates within human information processing system
  - e.g. STM limits etc.
- largely applied to problem solving in well-defined areas
  - e.g. puzzles rather than knowledge intensive areas

Problem solving (cont.)

- Analogy
  - analogical mapping:
    - novel problems in new domain?
    - use knowledge of similar problem from similar domain
  - analogical mapping difficult if domains are semantically different

- Skill acquisition
  - skilled activity characterized by chunking
    - lot of information is chunked to optimize STM
  - conceptual rather than superficial grouping of problems
  - information is structured more effectively

Psychology and the Design of Interactive System

- Some direct applications
  - e.g. blue acuity is poor
    $\Rightarrow$ blue should not be used for important detail

- However, correct application generally requires understanding of context in psychology, and an understanding of particular experimental conditions

- A lot of knowledge has been distilled in
  - guidelines
  - cognitive models
  - experimental and analytic evaluation techniques

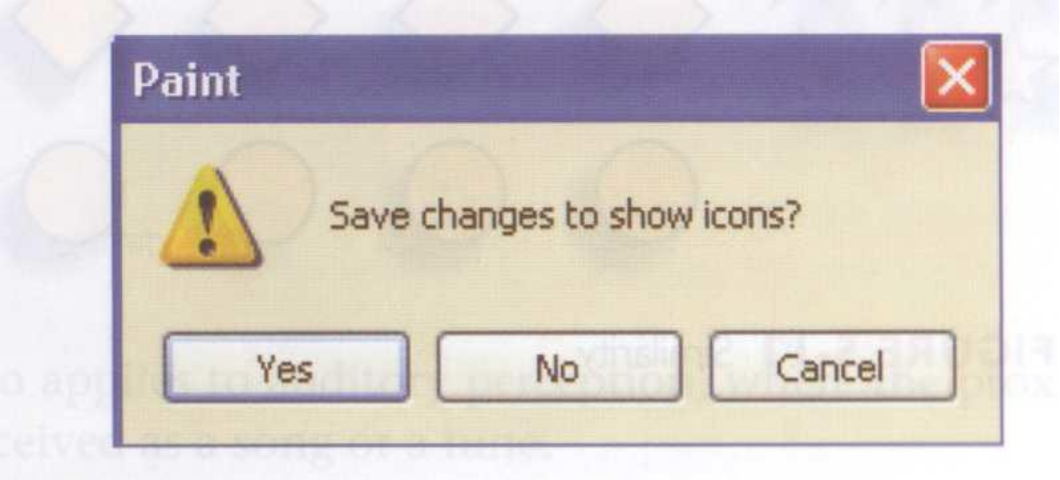
FALSE
true
True or False©1997 Scott Kim

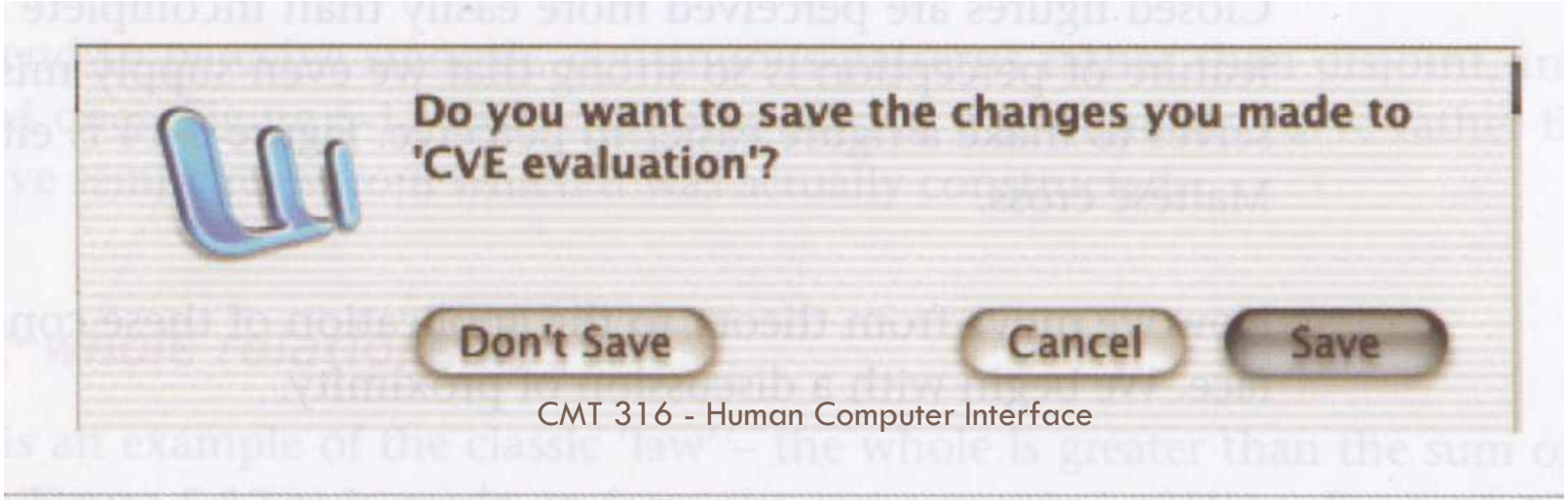
Lab Exercise:

- Using Gestalt Principle
  - In GUI (Graphical User Interface)
  - In design

Using Gestalt in GUI: proximity

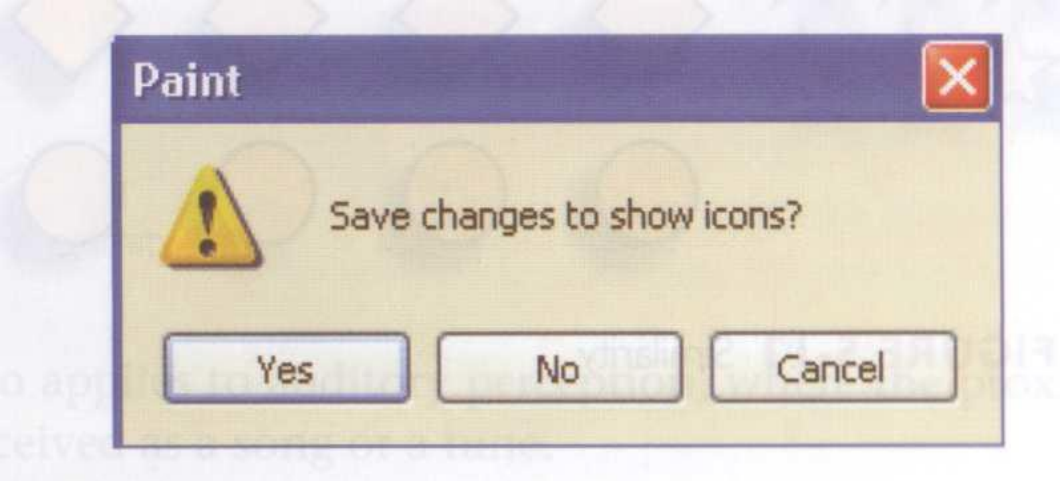
- Which is recommended?





Using Gestalt in GUI: proximity

- Not used



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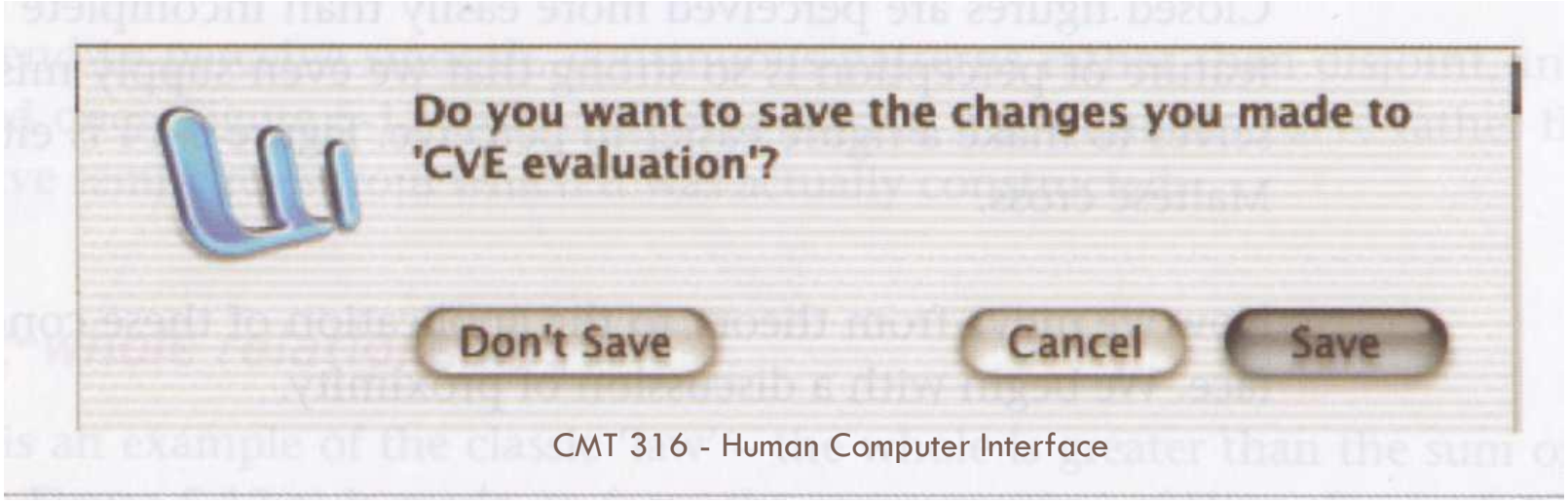
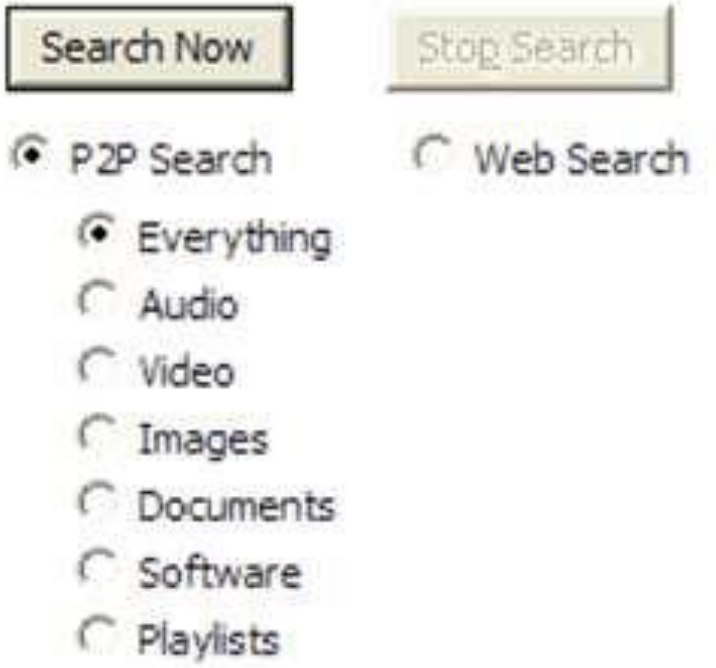


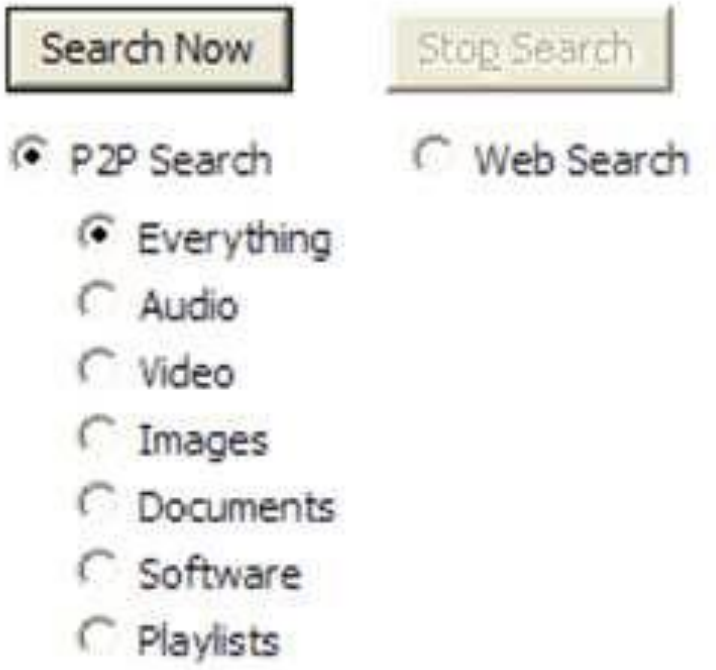
Using Gestalt in GUI: proximity

- Used



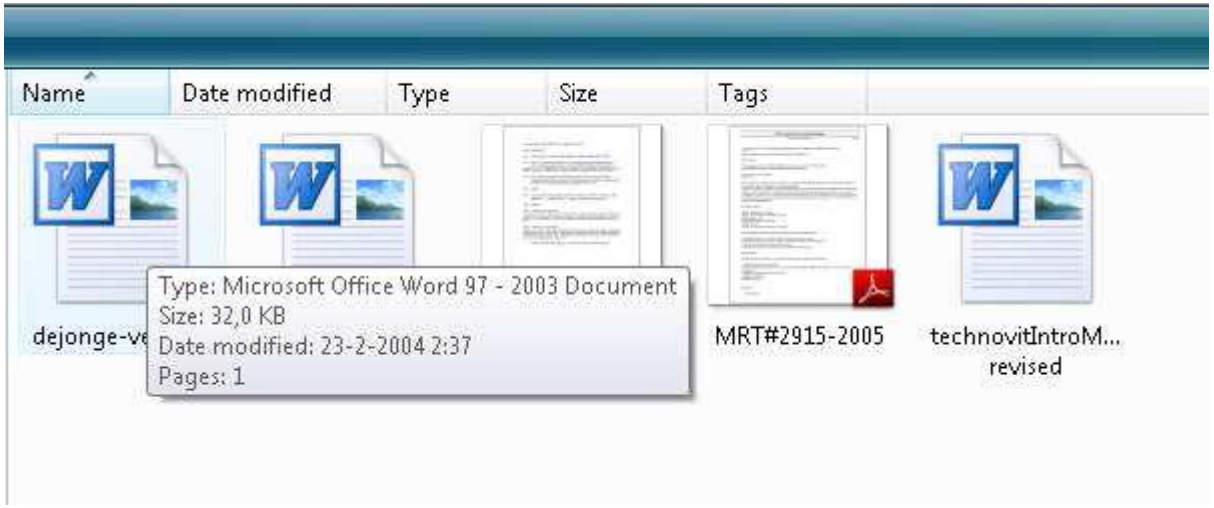
Using Gestalt in GUI: proximity

- Used (Enabling the current activity)



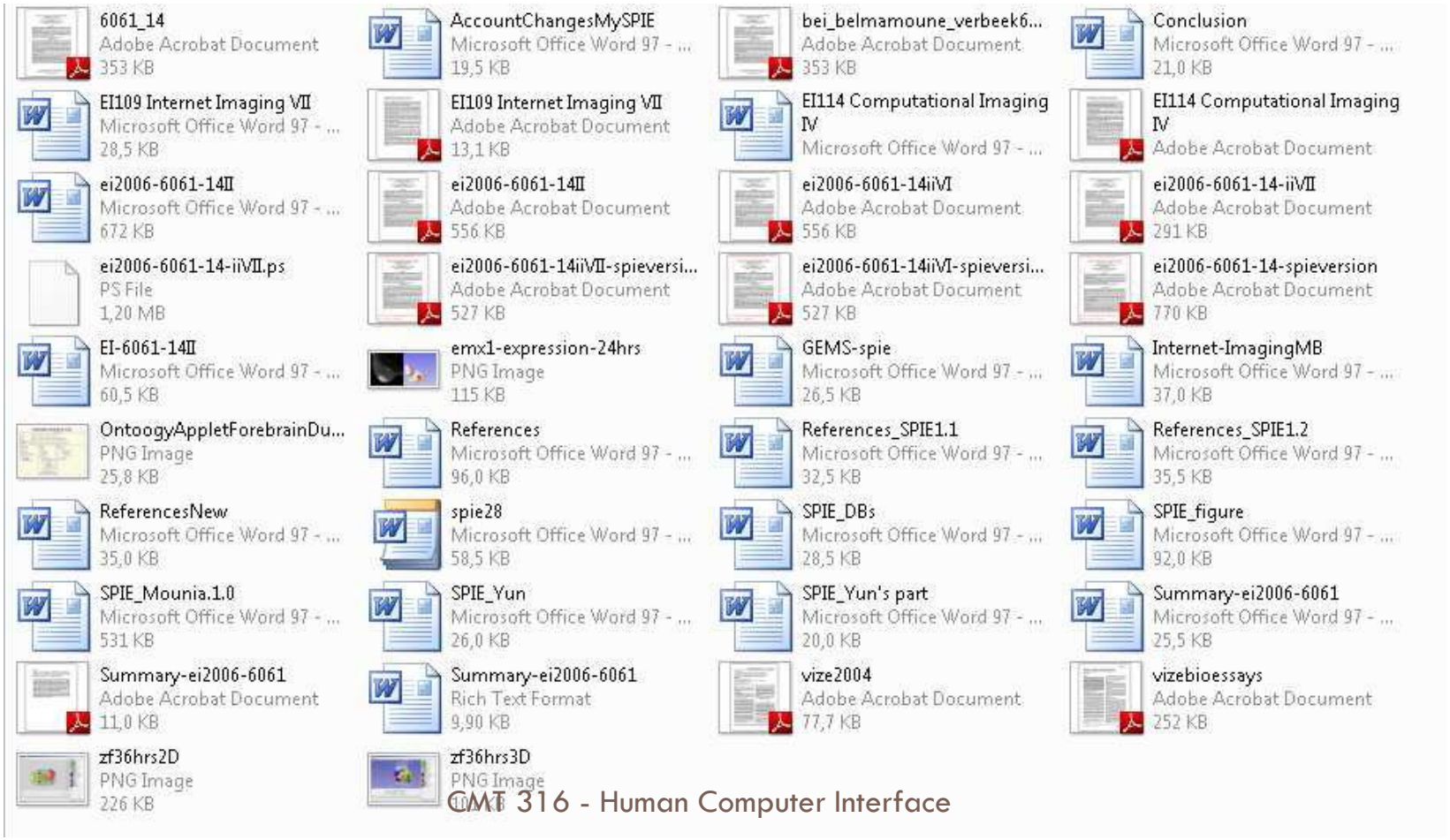
Using Gestalt in GUI: Similarity

- Used



Using Gestalt in GUI: Similarity

- Used

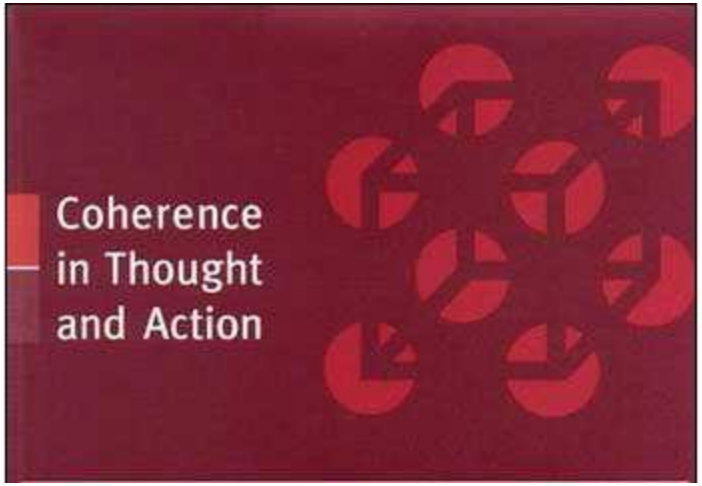


Gestalt in Design: Figure and Ground

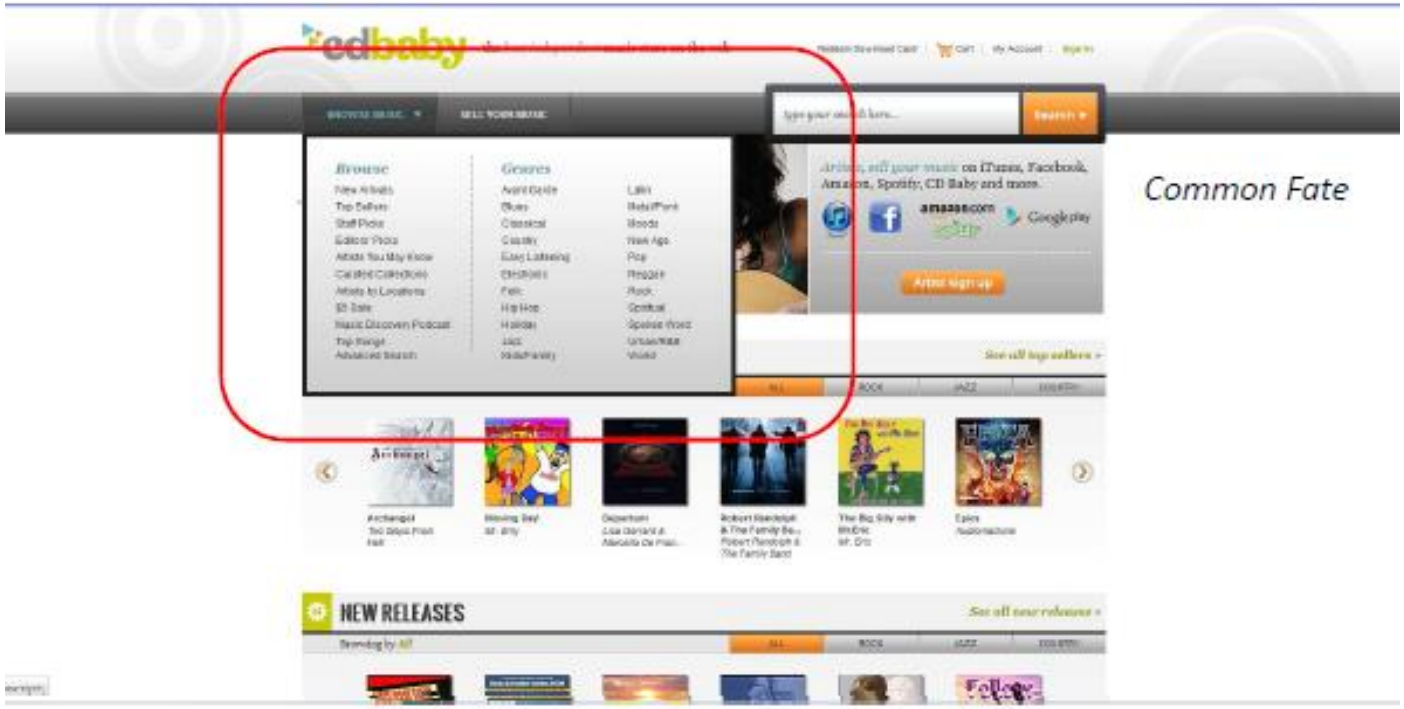
- Prominent role of the figure with respect to figure-ground
- Logo makers use this effect.

Gestalt in Design: Closure

- We see more than actually is presented
- We produce a closure to known artifacts



Website Design - Gestalt



Case Study-ATM

- Researchers + banks expected ATMs to be intuitively easy to use
- Testing among senior citizens found only 20% correct operation
- Senior citizens often put off by ATMs they find the machines
  - Complicated
  - Inconvenient

Problems with ATMs

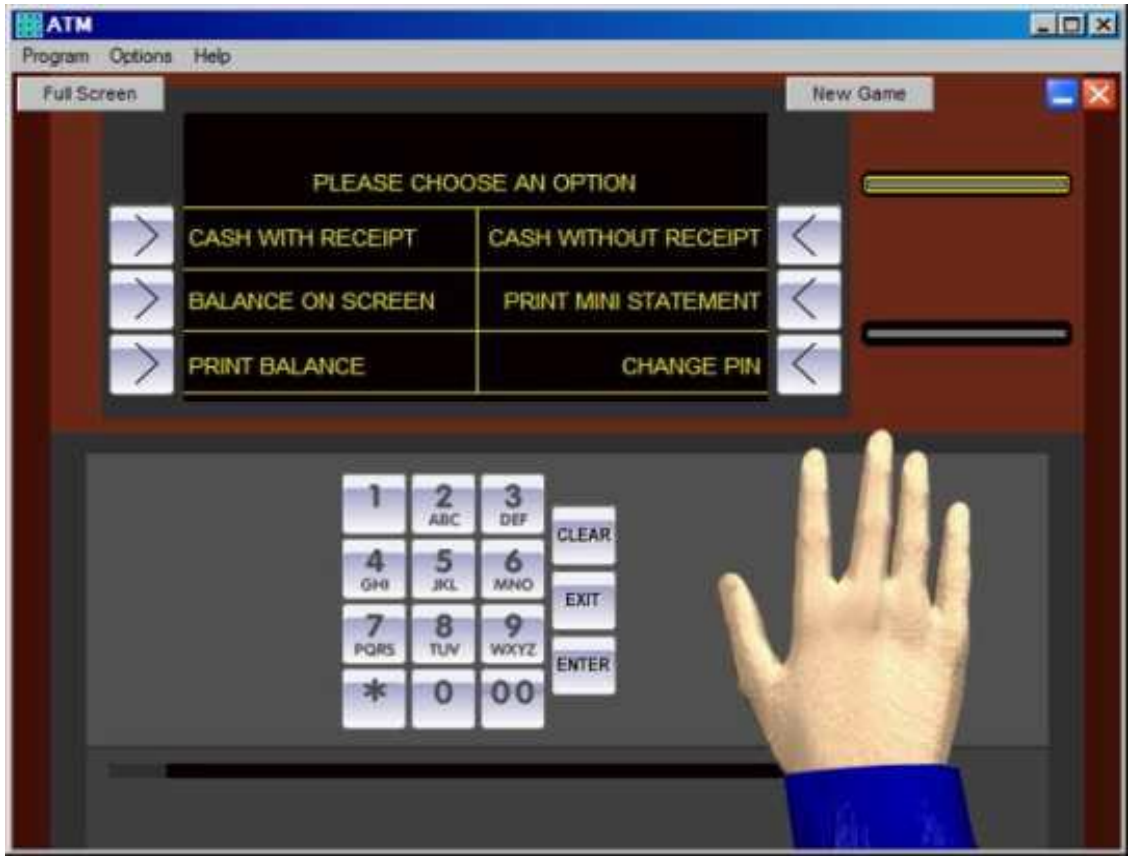
- Buttons that did not line up with commands

- Screens hard to read in the glare of daylight

- Sometimes confusing menu choices

Suggestions:

- For designers
  - Simpler on-screen instructions
  - More undo buttons
- For users
  - The user has to approach bank for providing sufficient training to use ATMs
  - Bank should also ready to serve the customers who are in need

For designers: Improvement with new technology

For users: Learn how to use it